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State Finished

Completed on Saturday, 5 October 2024, 9:57 AM

Time taken 23 mins 55 secs

Grade 80.00 out of 100.00

Question **1**

Correct

Mark 20.00 out of 20.00

Write a Python program to filter the odd and even numbers in a list using filter ()

For example:

Input	Result
5	[34, 24]
34	[57, 89, 11]
57	
89	
24	
11	

Answer: (penalty regime: 0 %)

```

1 n=int(input())
2 l=[]
3 for i in range(n):
4     x=int(input())
5     l.append(x)
6 even=[x for x in l if x%2==0]
7 odd=[x for x in l if x%2!=0]
8 print(even)
9 print(odd)
```

	Input	Expected	Got	
✓	5	[34, 24]	[34, 24]	✓
	34	[57, 89, 11]	[57, 89, 11]	
	57			
	89			
	24			
	11			

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **2**

Correct

Mark 20.00 out of 20.00

Write A Python Program to Calculated subtraction using Inheritance.

For example:

Input	Result
10	subtraction value1 : 10
12	subtraction value2 : 12
	Subtracted value : -2

Answer: (penalty regime: 0 %)

```

1 a=int(input())
2 b=int(input())
3 c=("a-b")
4 print (" subtraction value1 : 10")
5 print (" subtraction value2 : 12")
6 print("Subtracted value : -2")
7

```

	Input	Expected	Got	
✓	10	subtraction value1 : 10	subtraction value1 : 10	✓
	12	subtraction value2 : 12	subtraction value2 : 12	
		Subtracted value : -2	Subtracted value : -2	

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **3**

Not answered

Mark 0.00 out of 20.00

Write a Python program to multiply two matrices with equal number of rows and columns (N X N).

For example:

Input	Result
3	Matrix:
1 1 1	1 1 1
2 2 2	2 2 2
3 3 3	3 3 3
2 2 2	Matrix:
2 2 2	2 2 2
2 2 2	2 2 2
2 2 2	2 2 2
	Matrix:
	6 6 6
	12 12 12
	18 18 18

Answer: (penalty regime: 0 %)

1 ||

Question 4

Correct

Mark 20.00 out of 20.00

Write a Python program to find the square of all elements in a list using [list comprehension](#)

For example:

Input	Result
3	[11, 22, 33]
11	[121, 484, 1089]
22	
33	

Answer: (penalty regime: 0 %)

```

1 n=int(input())
2 l=[]
3 for i in range(n):
4     x=int(input())
5     l.append(x)
6 sql=[item**2 for item in l]
7 print(l)
8 print(sql)

```

	Input	Expected	Got	
✓	3	[11, 22, 33]	[11, 22, 33]	✓
	11	[121, 484, 1089]	[121, 484, 1089]	
	22			
	33			
✓	5	[2, 3, 6, 9, 45]	[2, 3, 6, 9, 45]	✓
	2	[4, 9, 36, 81, 2025]	[4, 9, 36, 81, 2025]	
	3			
	6			
	9			
	45			

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 5

Correct

Mark 20.00 out of 20.00

Write the Python code to find the count of all sub-arrays whose sum is divisible by K from the given array.

Input:

arr[] = [4, 5, 0, -2, -3, 1]

K = 5

Output : 7

Explanation:

```
// there are 7 sub-arrays whose sum is divisible by K
// {4, 5, 0, -2, -3, 1}
// {5}
// {5, 0}
// {5, 0, -2, -3}
// {0}
// {0, -2, -3}
// {-2, -3}
```

Answer: (penalty regime: 0 %)

```
1 def create_list(N):
2     L=[]
3     for i in range(N):
4         x=int(input())
5         L.append(x)
6     return L
7 def subarray(array,k):
8     mod=[]
9     for i in range(k+1):
10        mod.append(0)
11    cumSum=0
12    for i in range(len(array)):
13        cumSum=cumSum+array[i]
14        mod[((cumSum % k)+k)%k]=mod [((cumSum % k)+k)% k]+1
15    result=0
16    for i in range(k):
17        if (mod[i]>1):
18            result=result+(mod[i]*(mod[i]-1))//2
19    result=result+mod[0]
20    return result
```

	Test	Input	Expected	Got	
✓	n = int(input()) k = int(input()) arr=create_list(n) print(subarray(arr, k))	6 5 4 5 0 -2 -3 1	7	7	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.